

How Open Source Technologies Accelerate **Innovation**

Open source hardware and software are the new version of a phenomenon that's been with us since the first wandering bards added their own words to the melody of a well-loved folk tune. Open source tools feed the dreams of amateurs and entrepreneurs alike from a deep well of well-documented designs, which are constantly being refined by the communities that use them.

A growing number of traditional businesses and corporations understand that embracing them as an integral part of their business strategy may be their best hope for surviving the disruption and disintermediation which open source technologies often bring to tightly-held markets.



At first glance, the idea of freely-shared intellectual property (IP) would appear to contradict the fundamental principles of a competitive, market-driven economy. Open standards that enable interoperability between competing products are acknowledged as tools for building larger markets, but tightly-held IP and proprietary knowledge have been considered the key ingredients for innovation and success in technology based markets from the steam age onward (see box: 'Open Standards vs Open Source' for details).

This has begun to change over the past two or three decades, as the growing complexity of many technologies has given rise to several trends that threatened to slow the pace of innovation and market evolution:

- Many products are built upon layers of basic hardware and software functionality that are so complex it

would be impractical or impossible for most companies to develop themselves.

- Likewise, the resources needed to detect and fix the bugs in those technologies, or update them to meet evolving market requirements, are beyond the resources of all but the largest single company.
- Innovation is further discouraged by corporations that routinely create or acquire IP with the sole purpose of creating barriers to market entry for potential competitors.

These issues have helped drive up the cost of creating or licensing many essential IP elements to the point where they make many ventures less lucrative — or even impractical. Paradoxically, the stifling business conditions they created also served as the stimulus for the emergence and adoption of open, royalty-free technologies that are developed and are supported by the people who use them.

This emergence occurred in three distinct phases, beginning with open source software. The open hardware movement which followed somewhat later, adapted many of the concepts and licensing conventions developed over a decade by the software pioneers.

Recently, open source silicon platforms have begun to emerge as the third phase of the movement that is re-democratising innovation and powering the global tech economy.

A tale of two innovations

Phase 1: Linux

One of open source software's biggest successes is the Linux operating system, conceived by Linus Torvalds as a free alternative to AT&T's powerful but costly UNIX OS. Fittingly, Linux borrows many elements from Richard Stallman's earlier GNU project. Within a few years of its initial release in 1991, Linux had attracted a large,